

Dense Step-Level Rewards via Process Reward Models for Mathematical Reasoning

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Introduction & Motivation

- The standard RLHF (Reinforcement Learning from Human Feedback) system applies a single outcome reward at solution termination.
- This approach does not provide a signal on how a model reasoned, only whether it got the right answer, thus creating a sparse reward framework.
- In multi-step mathematical reasoning tasks, early errors can propagate through the chain and there is no way to correct this as the reward is awarded only at the end of the chain.
- Another drawback of this approach is that the sparse rewards cannot isolate where exactly reasoning fails.
- The solution proposed is a dense intermediate reward framework where a **Process Reward Model (PRM)** is used with a step-level scorer that is trained on annotated reasoning chains.
- The PRM rewards are applied in two ways: **(1) Inference Time Re-ranking** (Best-of- N) and **(2) Training Time Preference Signal** (DPO pair-wise generation).

Background & Prior Work

Sparse outcome based reward : These rewards are assigned only once at the end of the full solution. The issues with the sparse rewards are:

- Credit assignment is delayed especially across long reasoning tasks.
- No quantitative method of distinguishing correct answers from correctly reasoned ones.
- The gradient during RL training shows high-variance.

Dense process based rewards: A scalar reward is assigned at each reasoning step s_t , not just at the end.

$$r_t = \text{PRM}(x, s_1, \dots, s_t) \in [0, 1]$$

Prior work & how this builds on it:

- Lightman et al. [1]: Showed human step-level annotations improve verifier quality but are expensive to collect.
- Wang et al. [2]: Introduced Math-Shepherd for automated step annotation.
- Setlur et al. [3]: Process verifiers but focused on outcome-supervised training. This work applies step-level rewards to both inference-time re-ranking and DPO pair generation.

Setup & Experiments

Datasets

- GSM8K**: Consisting of 8.5k+ grade-school math word problems requiring reasoning. 80/20 train-test split.
- Math-Shepherd**: Consists of 400K+ math problems with step-level human annotations.

Baseline: Supervised Fine-Tuning (SFT)

- Using prior work on Qwen2.5-7B-Instruct with LoRA, an SFT was used as the baseline methodology.
- It was fine-tuned on GSM8K train for 3 epochs where the loss dropped from 1.43 to ~ 0.13 . Token accuracy also stabilized at 96%
- Greedy decoding on the test set showed an accuracy of **73.54%**

Goal Metric

The goal was to improve **test accuracy on GSM8K** over the established SFT baseline using a PRM based dense reward framework.

The secondary metric measured was the **AUC-ROC** of the PRM on a test set of the Math-Shepherd step labels. This will be used to measure intrinsic scorer quality.

PRM Architecture & Training

Architecture: The PRM was built using Qwen2.5-7B-Instruct along with a frozen SFT LoRA adapter. A step separator token **<step>** was used and a scalar reward head $\mathbf{W} \in \mathbb{R}^{3584 \times 1}$ was applied at each step-separator token position:

$$\hat{r}_t = \sigma(\mathbf{W}^\top h_{\text{sep}_t})$$

Training objective: Binary cross-entropy was the loss function used over all step labels:

$$\mathcal{L} = - \sum_t [y_t \log \hat{r}_t + (1 - y_t) \log(1 - \hat{r}_t)]$$

Best-of- N Re-ranking (Inference pipeline)

The PRM was used as an inference-time verifier thus requiring no additional training. The inference pipeline can be best depicted as:

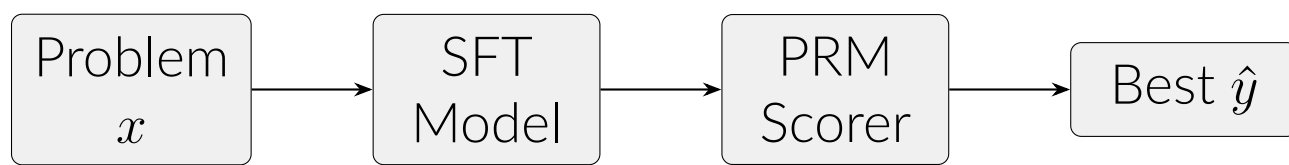


Figure 1. Best-of- N inference pipeline.

The procedure followed in order to select the best solution was:

- Sample $N(=8)$ solutions from SFT
- Score each: $\text{score}(y^{(i)}) = \frac{1}{T_i} \sum_{t=1}^{T_i} \hat{r}_t^{(i)}$
- Select $\hat{y} = \arg \max_i \text{score}(y^{(i)})$

DPO with PRM-Guided Pair Generation

For 500 problems from the GSM8K train problems $N=4$ solutions were sampled and scored by the PRM in order to form preference pairs. The following 3 strategies were used:

- Mixed outcomes**: Among the N samples, the highest PRM scoring, which was right with the most accurate reasoning, were chosen and the lowest scoring PRM scores were rejected. This strategy provided the strongest preference signals to the PRM.
- All correct**: All N samples had to be correct and the sample with the highest PRM score, best reasoning process, was selected while the lowest scoring one was rejected.
- All incorrect**: All N samples were incorrect and ranked from highest to lowest in PRM scores. If the score gap was < 0.05 , then the pair was skipped as the PRM could barely distinguish among their reasoning process.

Result: Only **64 pairs** emerged from 500 problems since the SFT model had a 91% accuracy, leaving almost no problems where meaningful contrasts could be formed. When DPO was trained on these 64 pairs overfitting was immediately observed gaining only **+0.15 pp** on the test set.

Results & Discussion

The primary metric (accuracy on the GSM8K test set consisting of 1,319 problems), was measured across the SFT (Baseline), the DPO fine-tuned solution and the Best of N solution, with $N=8$.

Method	Accuracy	Correct (1,319 problems)
SFT Greedy (baseline)	73.54%	970
DPO Fine-tuned	73.69%	972
Best-of-N + PRM ($N=8$)	86.58%	1142

Table 1. GSM8K test set accuracy (1,319 examples).

In order to measure the scorer's reliability, the PRM was evaluated on held-out step labels:

PRM Metric	Value
AUC-ROC	0.9287
Step Accuracy ($\tau=0.5$)	0.8463
Steps evaluated	10,150

Table 2. PRM on held-out Math-Shepherd steps (ckpt-12200).

- Best-of-8 result**: Showing a +13.04 pp over SFT with zero extra training, the Best of N ranking strategy showed tremendous promise. The PRM scores, with an AUC-ROC of 0.93, are consistent enough to pick the best candidate solution reliably.
- DPO was limited by the lack of Mixed outcome problems**: With only 64 pairs emerging from the 500 problems, DPO caused overfitting. Yet test accuracy improved by only +0.15 pp.
- Key insight**: The dense process rewards framework initially proposed proves to be effective at inference time especially when the base model (Greedy SFT) is strong. DPO pair generation does improve test set accuracy but requires harder, OOD problems to produce meaningful preference signal.

Planned Remaining Work

- Apply the Best-of- N and DPO framework on datasets with more complex problems, like MATH, AIME. Especially where SFT does not saturate training problems.
- Explore PPO with PRM as an online dense reward function, enabling RL fine-tuning
- Generate DPO pairs from held-out or harder problems and also introduce OOD problems to gain a better understanding of its ability to quantify reasoning.

References

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